ELECTROSTATIC ARTIFICIAL MUSCLE: COMPACT, HIGH-POWER LINEAR ACTUATORS WITH MULTIPLE-LAYER STRUCTURES

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ABSTRACT

Two types of compact, high-power, electrostatically driven actuators with multiple-layer structure, electrostatic artificial muscles, are developed. The actuators have 40 or 50 pairs of sheets interleaved together and enclosed in ready-to-use packages filled with a dielectric liquid. Sliding forces are generated on the surface of each film and are combined into a net force at the bundled edges of the sheets. The first type, a pulse-drive induction artificial muscle, which utilizes induced charges on highly resistive films, generated 8N propulsive force and 0.5W mechanical power with 110g its own mass. The second type, ac-drive variable-capacitance artificial muscle that weighs 3.6kg produced huge output, 310N, 19W. These actuators will be used in wide range of application fields scaling from micro manipulators to massive mobile robots.

INTRODUCTION

As the size of an electrostatic actuator decreases, its force / weight ratio increases. By linking a large number small actuators in parallel so that their individual force outputs are aggregated, a single compact, high-power actuator, called "Electrostatic Artificial Muscle", can be obtained. Recently, with this concept, some researchers have started the basic studies for large-scale electrostatic actuators with small elements [1,2,3]. In order to realize the artificial muscle, we have developed three types of linear actuators with multiple-layer structures [4-7]. A single-layer actuator consists of a pair of plastic film sheets, one slider and one stator. When many are stacked into a multiple-layer structure, a large total force is generated.

Table 1 shows three types of operation principles that have been previously presented, and categorizes them according to slider structures and excitation profiles. A pulse-drive induction actuator has the simplest structure, and is easiest to fabricate. A pulse-drive variable-capacitance actuator has a better power / weight ratio than the induction type. However, its power conversion efficiency is not satisfactory because the pulse voltages cannot be generated by a regenerative power source such as an inverter circuit. Conversely, an ac-drive variable-capacitance type provides better efficiency and achieves the highest power / weight ratio. A power to stator and slider weight ratio as high as 230W/kg, comparable to that of human muscle, has been achieved for a single-layer actuator.

In principle, a muscle-like compact and high-power actuator can be built using formerly developed single-layer actuators by linking them in parallel. In practice, however, parts unrelated to force generation such as frame, casing, and pins connecting the layers increase the total weight of the actuator; and consequently, their power to weight ratio is degraded. Furthermore, because the formerly developed apparatus were of experimental nature, they were not convenient for real applications such as robotic arms.

Table 1 Various Operation Principles

	Slider Structure	Excitation Profile
Pulse-Drive Induction	Resistor Coat	Pulse
Pulse-Drive Variable-Capacitance	Electrodes	Pulse
AC-Drive Variable-Capacitance	Electrodes	AC

This paper reports two types of compact, well-designed and ready-to-use actuator packages employing pulse-drive induction and ac-drive variable-capacitance multiple-layer actuators. As a practical application, a 2-DOF robotic arm, powered by a 40-layer induction type artificial muscle, is also presented.

PULSE-DRIVE INDUCTION ARTIFICIAL MUSCLE†

As shown in Fig. 3, an electrostatic artificial muscle consists of two groups of sheets stacked and interleaved together. Sliding forces are generated on the surface of each film and are combined into a net force at the bundled edges of the sheets. This section introduces a compact artificial muscle which utilizes the induction type film actuator developed by the authors [4,5].

Operating Principle

Fig. 1 shows the structure and driving method of a fundamental layer of the artificial muscle. This actuator consists of two sheets: a stator film and a slider film. In the stator, fine three-phase electrodes are embedded in parallel in an insulating film. The slider has a high-resistor layer that has a non-zero but extremely small conductivity.

The driving procedure consists of three steps: initial charging, actuation, and recharging. Initial charging is required only at the first excitation of electrodes. Actuation and recharging are repeatedly executed to continue the sliding motion.

(1) *Initial charging:* Positive and negative voltages are applied on the stator electrodes as shown in Fig. 1(1). The potential field induces electric charges of opposite polarity

† This section is written by S. Egawa.

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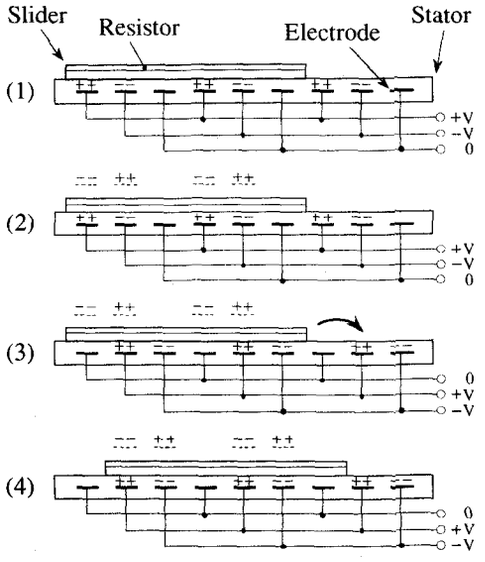


Fig. 1 Operating Principle

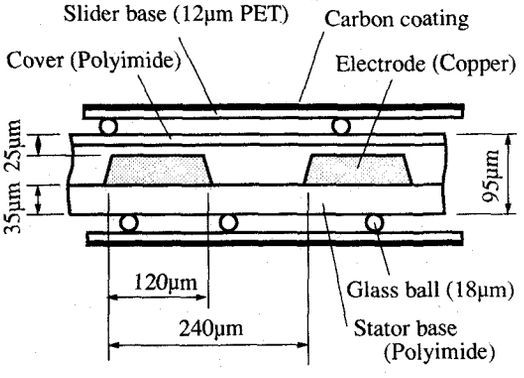


Fig. 2 Cross-section of Film Actuator

in the resistive layer on the slider (Fig. 1(2)). Because of the very low conductivity, it takes a certain amount of time to fully introduce charges on the resistive surface — almost one second for the currently fabricated actuator.

(2) *Actuation:* The applied voltage pattern is shifted as shown in Fig. 1(3). The charges on the stator electrodes move immediately according to the new potential. On the other hand, the slider charges stay at the same position because it takes relatively long time for the charges to traverse the resistive layer. The resultant electrostatic force between the charges drives the slider in the same direction as the voltage pattern is shifted (Fig. 1(4)).

(3) *Recharging:* After the slider rests at the next stable position, voltages are continuously applied to recharge the slider. The time for recharging is shorter than the initial charging time because most charges remain on the slider.

Fabrication of Film Actuators

Fig. 2 illustrates the film actuator construction. The stator is fabricated using conventional flexible printed circuit board (FPC) technology in which 240μm pitch electrodes are etched onto a 25μm polyimide backing film and subsequently covered with a polyimide film. The slider material is a 12μm PET film that has been coated with a carbon particle and polyurethane mixture which yields a

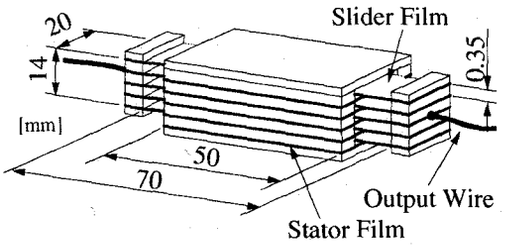


Fig. 3 Schematic View of 40-Layer Artificial Muscle

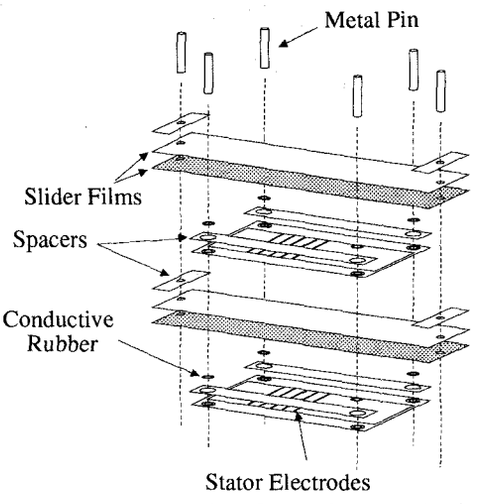


Fig. 4 Multiple-Layer Construction

$10^{14}\Omega/\square$ surface resistivity. The stator film is sandwiched by two slider films and a scattering of small glass spheres to reduce friction. Since electric discharge occurs in the air at relatively low voltage, the gap between the films is filled with a dielectric fluid (Fluorinert, 3M). By using the liquid, higher voltages can be used to get larger output forces.

Multiple Layer Construction and Packaging

Forty film actuators comprising forty stators and eighty sliders are stacked as shown in Fig. 3 and 4. Two slider films lay in each space between the stators, with their resistive surfaces facing each other. Spacers inserted between the films keep room for the sliders to move. Pins are used to fasten the films and spacers in both the stator and slider, respectively. Wires attached to the ends of the slider transmit the output force.

The pins are also used as electrical connection leads which distribute power from the driving circuit to each slider via conductive rubber rings around the pin and terminals on the sliders.

Fig. 5 shows a finished artificial muscle enclosed in a casing filled with the dielectric liquid. The portions of the slider extending out from the stator are enclosed by rigid plastic and the entire actuator is sealed with a heat-shrink tube. Transmission wires pass through pinholes in silicone rubber sheets affixed to the casing and take the actuator output out of the package. The total mass including the liquid is 110g. Structural members and liquid volume largely in excess of the amount needed between the film layers account for a large part of the mass. In fact, only 30% is directly related to force generation. Volume of ineffectual parts should be decreased in the future models.

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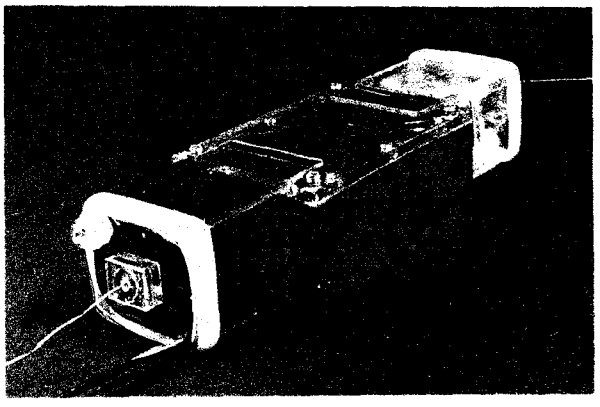


Fig. 5 40-Layer Artificial Muscle Package

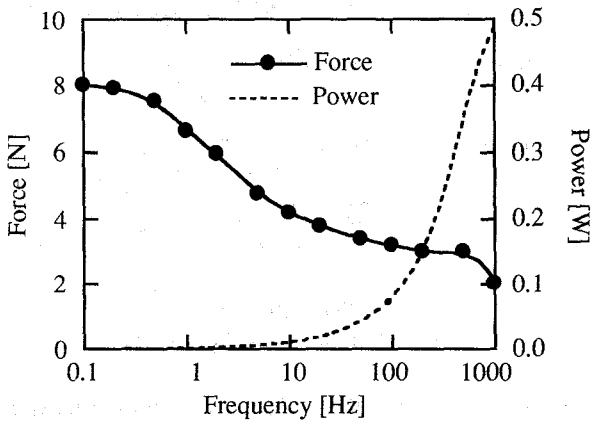


Fig. 6 Force and Power of 40 Layer Artificial Muscle

Experiments

Propulsive force was determined by finding the maximum weight that can be lifted. Fig. 6 shows the relation between force and driving pulse frequency obtained with the applied voltage of 800V, the limit level to avoid breakdown of the insulator in the stator. Generated power, calculated using the velocity estimated from the pulse rate, is included in the figure. As the frequency increases, the force decreases. The power comes to the maximum just prior to the point before the slider stops moving at 1000Hz.

The force of the 40 layer artificial muscle is 30 times as large as a test result with a single layer actuator. The reason why the force is not proportional to the number of layers will be that the charges on the pairs of sliders inserted in each gap between stators interfere each other.

Table 2 summarizes the performance of the artificial muscle. The ideal values for force and power density were estimated by neglecting the non-force generation related masses. The actual performance will approach these values, as the designs of structural parts are optimized.

Robotic Arm Driven by Artificial Muscle

As a model application of the artificial muscle, a simple two degrees of freedom robotic arm driven by two artificial muscles have been constructed. As shown in Fig. 7, the arm structure built with PMMA frames, whose total length is 390mm and weight including actuators is 285g, can carry 50g payload. An output wire coming out from the bottom of

Table 2 Induction Type Artificail Muscle Performance

Electrode pitch	240μm
Number of layers	40
Effective area	768cm^2
Mass	110g (34)*
Driving voltage	800V
Maximum force	8N
Maximum power	0.5W
Force / Weight	73N/kg (240)*
Power / Weight	5W/kg (15)*

* ideal values estimated by neglecting ineffectual masses

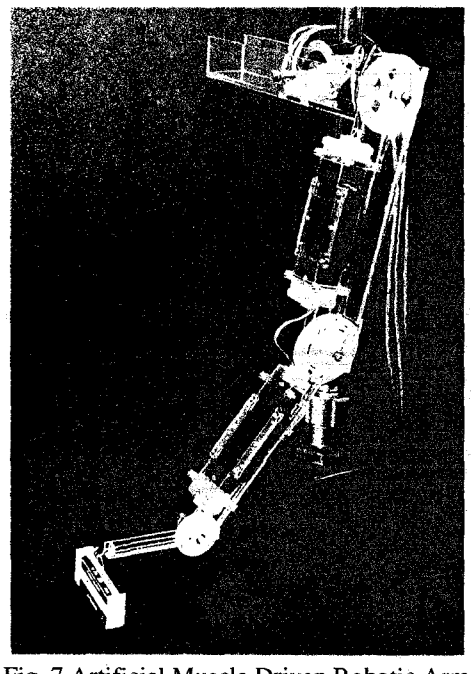


Fig. 7 Artificial Muscle Driven Robotic Arm

an actuator winds around two pulleys and returns to the actuator from the top. Because a preload is imposed on the wire loop, the actuator can rotate the joints for both directions. The forearm actuator has only 20 layers to reduce the actuator weight and decrease the load on the upper arm.

The joint angles are feedback controlled using potentiometers on the joints. Since the actuators move freely when the power is turned off, one can teach an arm path by leading the arm by hand.

AC-DRIVE VARIABLE-CAPACITANCE ARTIFICIAL MUSCLE

Operation Principle

Three possible kinds of operation methods have been developed [7]. This chapter describes the simplest and most convenient method: *single-frequency method.* Fig. 8 shows the schematic cross-sectional view of a primitive single-layer actuator and necessary wire-connections for the single-frequency method. The actuator consists of two elements: a slider and stator. In the both elements, 3-phase electrodes are embedded and connected to a 3-phase power source, with the slider order being the reverse of the stator order. Upon excitation, potential distributions traveling in the opposite

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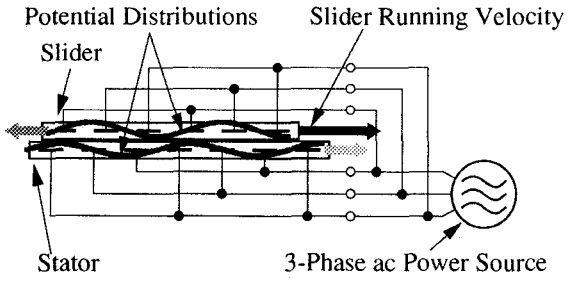


Fig. 8 Single Frequency Method

Excitation by a 3-phase ac power supply generates two traveling potential distributions, whose field interaction drives the slider.

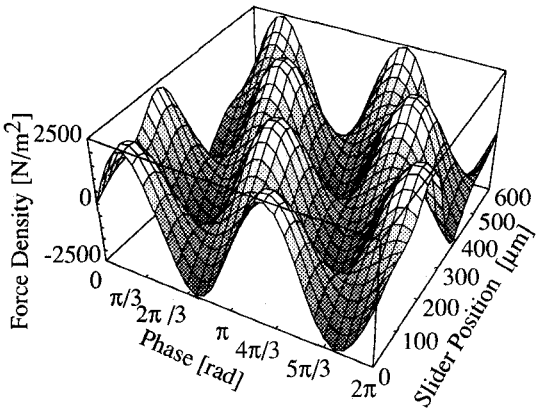


Fig. 9 Calculation For Force Generation

Voltage = 1400V r.m.s. Electrode Pitch = 200mm

directions along the slider and stator electrodes are generated as depicted by the thick lines in Fig. 8. A field interaction between the two distributions produces electrostatic force, dependent on the phase difference, that actuates the slider. The synchronized speed of the slider, u_a, is

$$u_a = \lambda \frac{\omega}{\pi} \qquad (1)$$

where ω is the angular frequency of the 3-phase ac and λ is the wavelength of the distributions, equivalent to three times the electrode pitch.

In practice, since the electrodes have a limited number of phases and are located discretely, the potential distributions around the electrodes are not completely sinusoidal, and their profiles change as they move. Therefore, the actuation force cannot be simply regarded as single-variable function of phase difference, but is instead a function of both slider position and phase of ac voltage. Fig. 9 is a numerical calculation of the relationship using surface charge method. Fig. 10 shows the force generated when the slider moves at a constant synchronized speed. Since the profiles of the distributions change as the slider moves, the actuation force fluctuates despite the synchronization. These fluctuations, called torque ripple, decrease controllability of the actuator. However, it is expected that decrease of this ripple will not be difficult by improving the voltage profile.

Fabrication

Both the slider and stator were fabricated in the same manner using a flexible printed circuit board technique. Fig.

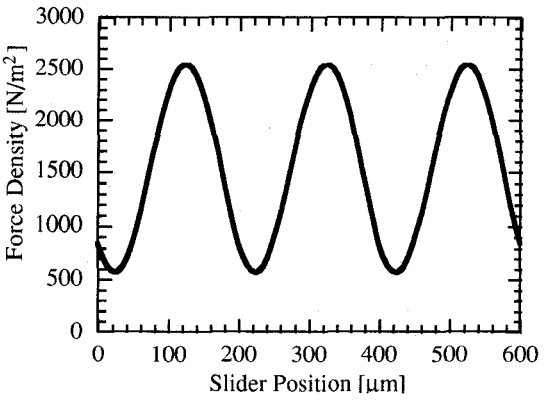


Fig. 10 Torque Ripple

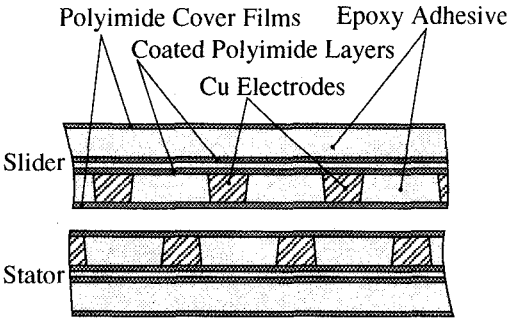


Fig. 11 Cross-Sectional View of a Slider and Stator

Polyimide thickness = 12.5μm, Cu thickness = 40μm, Total thickness of a slider or stator = 140μm. Thorugh-hole pads are not shown. Cu thickness exceeds original foil thickness due to plating process.

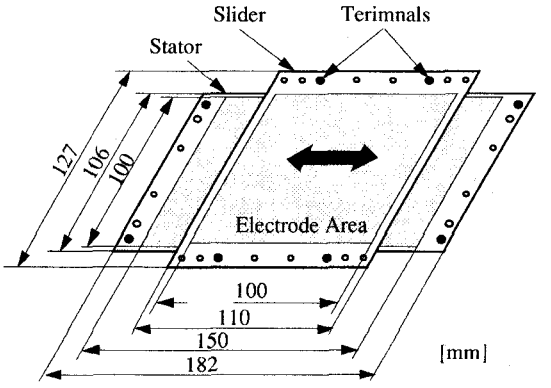


Fig. 12 Electrode and Terminal Layout

Effective intersection area between slider and stator is 100cm², irrespective of slider position.

11 is the cross-sectional view of a single slider and stator layer. The process for each sheet starts with two sheets of copper foil of 25μm thickness, onto which polyimide resin is coated. Their polyimide sides are bonded together with epoxy adhesive, effectively creating a single sheet which has copper film on the both sides. Next, interconnection bridges, which are required to afford the 3-phase structure of the electrodes, are then achieved by piercing through-holes into the sheet and depositing an additional copper layer on all surfaces by electroless plating. Etching is then used to form 200μm-pitch electrodes on one side and through-hole pads

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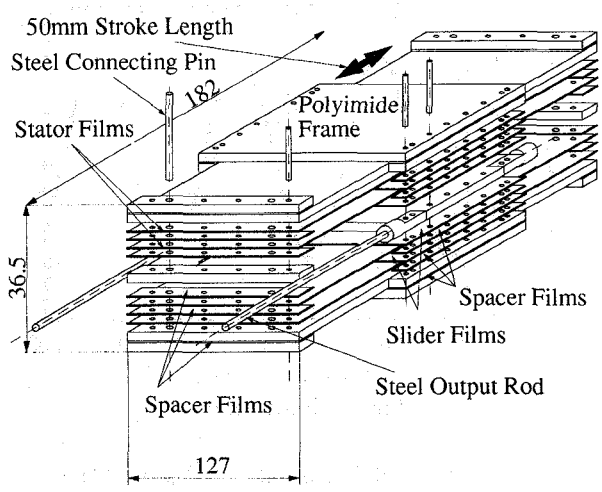


Fig. 13 Structure of 50-Layer Slider-Stator Unit

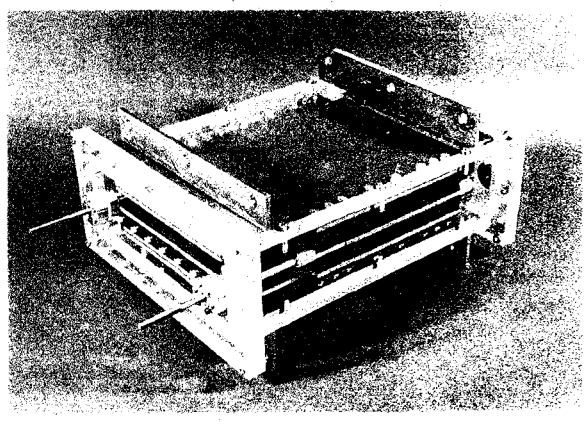


Fig. 14 50-layer AC-Drive Variable-Capacitance Actuator

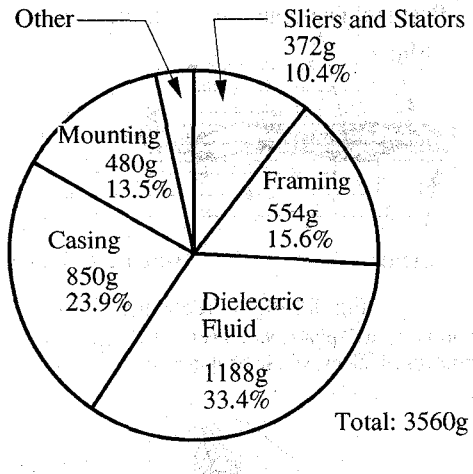


Fig. 15 Weight Budget of the Actuator

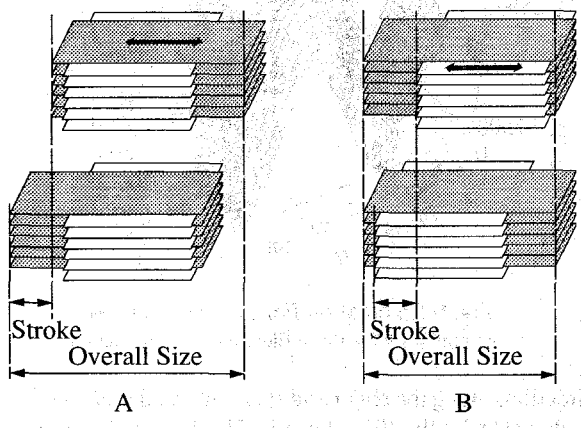


Fig. 16 Slider-Stator Configurations

A: Stator shorter than slider. B: Slider shorter than stator. Configuration B provides better space usage with the same stroke.

on the other. Finally 12.5μm thick polyimide films are adhered to the exposed copper to act as a protective insulating layer. Fig. 12 shows the layout of electrodes and their respective terminals in both the slider and stator sheets.

Fifty sliders and stators are stacked alternately to create a 50-layer slider-stator unit as shown in Fig. 13. Linkages between slider layers are made by eight steel pins of 3mm diameter. Similarly, linkages between stator layers are made by four pins of 4mm diameter. Between every layer, a polyimide spacer sheet with a thickness of 250μm is inserted in order to prevent the sliders and stators from squeezing each other. Finally, the alternating slider and stator layers are sandwiched by 3mm thick polyimide plastic sheets, which have tensile strengths of 190MPa, comparable to that of aluminum yet half the density (1.4 g/cm^3).

The entire slider-stator unit is enclosed by an acrylic plastic casing which is filled with dielectric fluid, in order to prevent dielectric breakdown. Two 4mm diameter steel connecting rods transfer the total working force of the slider-stator unit out of the casing. Fig. 14 shows a photograph of the 50-layer actuator. The total weight of the actuator including all the parts is 3.6kg. A breakdown of the actuator's weight budget is given in Fig. 15.

Optimized Slider Stator Configuration

In order to improve power / weight ratio of a multiple-layer actuator, a particular effort must be made to minimize the weight of non-force generating components. In this way, the actuator was developed to optimize space usage and reduce the weight of dielectric fluid.

As shown in Fig. 16, there are two configurations that can be implemented. In the first, the slider is longer than the stator. Conversely, in the second, the slider is shorter than the stator. It can be seen that the second configuration is preferable because it offers better space efficiency for an equivalent stroke length and decreases the amount of dielectric fluid.

Experimental Results

Fig. 17 is the relation between actuation force and the amplitude of the 3-phase ac voltage which is applied to the actuator electrodes. The resultant force is approximately proportional to the square of the voltage; and a maximum value of 310N, or 32kgf, is achieved by an excitation of 1400V r.m.s.. The maximum value is limited by dielectric breakdown between two neighboring electrodes. When the applied voltage is lower than 300V, the actuator does not overcome 3N of frictional force and fails to operate.

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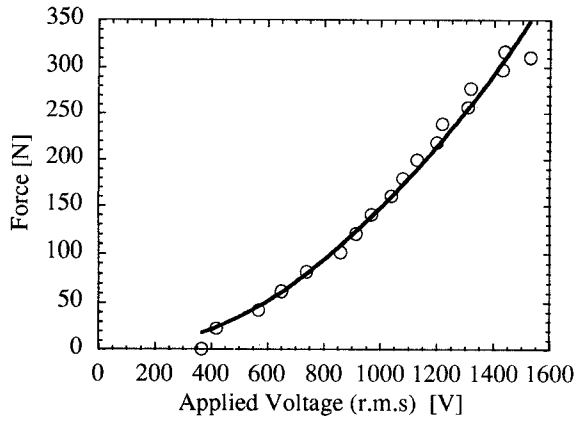


Fig. 17 Relation between Force and Applied Voltage
The voltage is that between two terminals of the 3-phase ac.

Output Force Maximization

Since the actuator generates a force proportional to the square of the applied voltage, the best way to increase output force is to improve dielectric breakdown strengths in the slider and stator. Especially, the strength of the adhesive material, which bonds the polyimide cover film to the Cu electrodes, is important since most breakdowns happen in the material.

It is expected that improving this strength by a factor of two will not be difficult. In fact, the slider and stator electrodes have been subjected to voltages in excess of 3000V for several minutes without incident in the lab. If such a high voltage becomes practicable for regular use, the actuator could generate more than 650N.

COMPARISON OF PRINCIPLES

This section compares four possible operation principle types: pulse-drive induction, ac-drive induction, pulse-drive variable-capacitance and ac-drive variable-capacitance.

Table 3 compares the principles from several points of view. Although fabrication of the two variable-capacitance actuators tends to be difficult (since all the electrodes must be kept in parallel), the best power generation can be achieved. In terms of both controllability and efficiency, the two ac-drive actuators are better than the two pulse-drive actuators because they have reduced torque ripples and regenerative inverter circuits can be employed to apply the sinusoidal voltages. With respect to driver simplicity, the two pulse-drive actuators are the best as they do not need complicated circuits and simple switches can be employed.

In any case, the ideal operating principle depends on the purpose of the actuator. For applications which require extremely high-power (i.e. robotic arms or legs), the ac-drive variable-capacitance type is the best. They can provide high muscle-like power with good efficiency. However, if more controllability is needed, an ac-drive induction actuator seems better. Its torque ripple is quite small, though it does not generate so much force as the ac-drive variable-capacitance type. Although pulse-drive types are not suitable for high-power application, their single-layer models are convenient for applications in which complicated drivers cannot be implemented.

Table 3 Comparison of Principles

	PDI	ACI	PDV	ACV
Ease of Fabrication	★★	★★	★	★
Power Generation	★	★	★★	★★★
Controllablility		★★		★
Efficiency		★★		★★
Driver Simplicity	★★	★	★★	★

PDI: Pulse-Drive Induction, PDV: Pulse-Drive Variable-capacitance
ACI: AC-drive Induction, ACV: AC-drive Variable-capacitance

CONCLUSIONS

Two types of electrostatic artificial muscle with multiple layered structure, pulse-drive induction type and ac-drive variable-capacitance type, are developed.

A 40 layer PDI muscle that weighs 110g is fabricated and produced 8N propulsive force and 0.5W mechanical power. Though the output, efficiency, and controllability of PDI are not so good as those of ACV, it is easier to fabricate. PDI actuators will be used with single or small number of layers in the application fields such as thin film transportation or micro manipulation, where compactness is more strongly required than power output.

ACV type generates very large output. The 50 layer, 3.6kg ACV muscle produced 310N propulsive force and 19W mechanical power. In near future, ACV muscle is expected to excel human muscles as well as conventional electromagnetic motors in force and power densities. ACV muscle will be used in applications where large power and lightweight actuators are needed, such as mobile robots.

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